Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/10/2014
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Project Title:	Material Ignition and Suppression Test (MIST) in Space Exploration Atmospheres		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Combustion science		
Joint Agency Name:		**TechPort:**	No
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Space Biology Element:	None		
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Zip Code:	94720-1740	**Congressional District:**	9
Comments:			
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Flight Program:			
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NOTE: Continuation of NNX10AE01G, Material Ignition and Suppression in Space Exploration Atmospheres. Central Objectives: The next generation of NASA spacecraft will use lower pressures and increased oxygen concentrations than current designs. Although there is insufficient knowledge about fire behavior of materials in these Space Exploration Atmospheres (SEA), recent research suggests that material ignitability is increased in SEA. These results imply that next generation spacecraft present a greater fire safety hazard than current designs. Although the most effective fire safety strategy is to prevent ignition, if ignition does occur, it is critical to rapidly extinguish the fire. Therefore, an experimental investigation into ease of ignition and fire suppression characteristics of materials under SEA is proposed.

Methods/techniques: The proposed experimental apparatus is simple and easily incorporated in the Combustion

Task Description:	Integrated Rack on the International Space Station. It consists of a small-scale cylindrical flow duct that has fuel samples attached to its inside wall. A cylindrical radiant heater is located at the center of the duct. An oxidizer gas flows at down the length of the tube. Electrical igniters are placed downstream of each fuel sample. Time to ignition and the time between introduction of suppression agent and extinction are measured as a function of environmental conditions. The normal gravity and microgravity experimental programs will be supported by a comprehensive model of piloted ignition and extinction of materials in SEA. Microgravity experiments will be used to validate the model, which in turn will be used to predict the fire behavior of different combustible materials in SEA. Perceived Significance: Currently, there exists no testing methodology specifically designed to determine the fire hazards of materials under those conditions. The proposed work will fill that void and provide additional information about the effect of SEA on the flammability and suppression of materials. It will also provide guidance to interpret and extend current NASA testing procedures to SEA.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The project is intended to enhance the ability to predict ignition, suppression, and spread characteristics of materials in their end use configuration. This approach is called performance based design in terrestrial fire protection engineering. Computational support for this approach requires detailed information about material thermal and combustion properties. Our work on MIST will use experimental data to validate computer models attempting to predict fire behavior using material property estimation techniques. These techniques are currently a focus of research in fire protection engineering.
Task Progress:	The Material Ignition and Suppression Test (MIST) project was developed to help in better understanding the ignition and suppression of radiatively heated, piloted, solid fuel samples in Space Exploration Atmospheres (SEA). MIST project was a joint experimental and numerical investigation of the flammability of materials under non-standard atmospheric conditions, with the eventual goal of helping in the development of better flammability tests. Current experimental investigations surround the the piloted ignition of cylindrical samples of Poly(methyl-methacrylate) (PMMA) under an externally applied radiant heat flux. Clear and black PMMA rods are tested in a longitudinal orientation in a natural convection air flow. This work expands upon previous investigations of ignition phenomena which were concerned with the piloted ignition characteristics of flat samples of PMMA or similar thermoplastics. The primary novel aspect of this experimental investigation is the cylindrical symmetry of the samples and the associated curvature of the fluid, thermal, and species boundary layers which act to change the thermo-chemical characteristics of the sample heating and ignition. Thus far, there seem to be (1) slightly longer ignition delay times when compared to similar conditions for flat samples, and (2) similar fuel mass flux at ignition. Both of these phenomena can be attributed at least in part to the symmetrical boundary condition at the center of the cylindrical samples, and the thinner boundary layer present in the gas phase -- due to the curvature of the sample surface. It is proposed that the thinner boundary layer may result in greater convective cooling of the sample, thus increasing the time to ignition, without significantly affecting the critical mass flux necessary for ignition. Modeling has been carried out in conjunction with the above experimental work to simulate the MIST experiments. The 2-D antisymmetric computational model mimics the geometry of the current experiment using a long antisymmetric fuel rod in a duct of similar dimensions to that in the current iteration of the experiment. Fire Dynamics Simulator (FDS) version 6.0.1, developed by NIST, in its Direct Numerical Simulation (DNS) mode was employed. The model simultaneously considers the processes in the solid and in the gas phases. The solid phase decomposition is modeled with a single step global Arrhenius reaction rate. Oxidative pyrolysis is not considered and the in-depth formed pyrolyzate is assumed to flow unrestricted through the solid combustible. Additionally, due to limitations in FDS, conduction in the solid is strictly radial. Gas phase kinetics are modeled with a single-step second order Arrhenius reaction rate. Although the current model simulates the current set of experiments quite well, it may be necessary to validate the predictive capabilities of the model and quantify how well the current values for physical parameters are able to handle environmental changes and then implement any necessary modification as necessary. Thus far, the modeling results seem to indicate that buoyancy plays an important role via the Rayleigh-Taylor instability at normal gravity, particularly once the flame is established over the sample surface. However, the Rayleigh-Taylor instability disappears for zero gravity conditions and the flame shows a distinctly round, more uniform shape in that case. Furthermore, ignition occurs significantly earlier in 0-g than in 1-g conditions. The computational model is moving towards becoming useful predictive tools for the study of ignition and flame spread phenomena. Eventually, the goal will be to have validated the model using both 1-g and 0-g experimental data and then be able to extend the model to the predictive realm with better confidence as to its output. Both the experimental and computational results will hopefully assist in the future development of better fire safety tests for employ in manned space capsules.
Bibliography Type:	Description: (Last Updated: 02/15/2026)
Abstracts for Journals and Proceedings	Link S, Fernandez-Pello C, Urban D, Ruff G. "An Experimental Investigation of the Piloted Ignition of Cylindrical PMMA Under Axial Flow." Presented at the Fall 2013 Western States Section/The Combustion Institute Technical Meeting, Fort Collins, CO, October 7-8, 2013. Fall 2013 Western States Section/The Combustion Institute Technical Meeting, Fort Collins, CO, October 7-8, 2013. http://www.eecl.colostate.edu/combustioninstitute/wssci-fall2013-program.pdf , Oct-2013
Articles in Peer-reviewed Journals	Fereres S, Fernandez-Pello C, Urban D, Ruff G. "Identifying the Roles of Reduced Gravity and Pressure on the Piloted Ignition of Solid Combustibles." Combustion and Flame. Submitted, as of April 2014. Expected publication October 2014. , Apr-2014